

GridAI : GReen Itinerant Distributed AI

Follow the sun and the wind, making AI workloads nomadic, the renewable way

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Abstract—The growth of AI workloads is on a collision course with the European energy transition: datacentre electricity demand is on course to double by the end of the decade, while substantial volumes of renewable energy are routinely curtailed across the continent because the existing electrical grid has not yet been sized to move them at the scale and pace required to reach where compute is concentrated. The dominant European response is the construction of transcontinental high-voltage direct-current corridors that physically move electrons to where compute is. This paper asks the dual question: under which conditions can the computation be moved to where the electrons are? We extend our prior feasibility-domain model for single-GPU workload migration along two dimensions that the prior work did not address, namely the temporal flexibility of the workload (deferrability) and the storage-based flexibility of the supporting infrastructure. The resulting framework, which we call *nomadic AI compute*, defines the space-time *nomadic envelope* of an eligible AI workload, articulates three explicitly-priced levers on the bit $\leftrightarrow$ electron $\leftrightarrow$ time hinge, and identifies a storage–compute equivalence principle under which a fraction of grid-scale flexibility can be built in silicon rather than in lithium. The framework applies to AI workloads with an inter-site-compatible algorithmic regime (single-GPU, federated, asynchronous, compressed, sharded, inference) and explicitly excludes fully synchronous data-parallel training of frontier-scale models, recovering the prior feasibility-domain model as the one-dimensional degenerate case. We describe the multi-scale green infrastructure design (nano-datacentres on low voltage, micro-datacentres on medium and high voltage), the technological stack (programmable data plane on SRv6 and IOAM substrates, multi-lever orchestrator, verifiable green-compute certification), and the Italian federated testbed (anchored on the PrinceLab cyber-physical platform and on the GARR national research backbone) on which the framework is being empirically validated against a Llama-class federated continual learning headline workload. A companion document accompanying the paper articulates three concrete tracks for European collaboration: cross-national validation of the simulator, cross-deployment of the federated

testbed, and IETF standardisation around green-computing monitoring and certification.

Index Terms—Green computing, renewable energy, distributed AI, federated learning, workload migration, energy-aware orchestration, carbon-aware computing, programmable data plane, SRv6, IOAM, nomadic compute, AI-RAN, HVDC, demand response, grid flexibility, verifiable computing.

I. INTRODUCTION

The collision between the energy demand of artificial intelligence and the trajectory of the European energy transition is, at this point, no longer a forecast. Datacentre electricity demand is on course to double by the end of the decade under the pressure of AI workloads [1], while in 2024 alone the European electrical system curtailed 338 GWh of renewable generation in Italy [2] and a total of EUR 7.2 billion across seven countries (with Germany and Spain alone accounting for EUR 3.3 and 2.5 billion respectively) [3] because the existing transmission infrastructure cannot deliver the surplus to the consumption centres. The dominant European response to this asymmetry is the construction of transcontinental high-voltage direct-current corridors that physically transport electrons from generation sites to compute sites [3]. This direction of travel is necessary and will be pursued; what has been largely absent from the policy and engineering conversation is the dual question of whether, and at which cost, the computation could be moved to where the electrons already are.

A first quantitative engagement with this question was provided in our prior work [4], which formalised a feasibility-domain model for the spatial migration of single-GPU AI workloads under renewable-surplus windows. The model showed that, for a large and operationally significant class of workloads, the binding constraint on physical migration is not

the energy cost of moving the state but the time required to do so within the window during which the renewable surplus is available. Two findings carried into the present paper: the energy break-even of a typical single-GPU migration is in the order of minutes, against renewable windows that last hours; and the feasibility partitions cleanly into a small number of regimes that align with the typical checkpoint sizes of contemporary single-GPU AI jobs.

This paper extends that prior work along the two dimensions that the feasibility-domain model did not address. The first dimension is the temporal flexibility of the workload itself: not every AI workload must run immediately, and the range of acceptable delays spans several orders of magnitude across the workloads that traverse a modern compute infrastructure. The second dimension is the storage-based flexibility of the supporting infrastructure: an energy-proximate compute site can be co-designed with local battery storage and with a buffer of deferrable workloads, and the resulting infrastructure has a richer set of operational levers than a site that can only absorb its renewable surplus instantaneously. Adding the two dimensions changes the design space of the renewable-native AI infrastructure in qualitative, not merely quantitative, ways.

The framework we develop, which we refer to as *nomadic AI compute*, makes three structural contributions. First, it defines the *nomadic envelope* of an AI workload as the space-time surface within which the workload can be executed while preserving its quality of service, with two physical and measurable dimensions (spatial mobility and temporal deferrability) that are today invisible to AI orchestration and that the framework makes explicit, controllable, and auditable. Second, it articulates three explicitly-priced levers on the bit $\leftrightarrow$ electron $\leftrightarrow$ time hinge, and it assembles them into a single cost functional that the orchestrator minimises under a common metric, allowing the trade-off between moving bits, moving electrons, and moving in time to be quantified rather than negotiated through three disjoint markets. Third, it identifies a *storage–compute equivalence principle* under which a renewable surplus can be absorbed either by charging a physical battery or by accelerating a deferrable workload from the backlog, with identical effect on the carbon balance and with explicit costs on either side; the principle makes the partition between lithium-side and silicon-side flexibility a controllable design variable rather than an implicit assumption.

The framework is not unconditional: it applies to AI workloads whose algorithmic regime is compatible with inter-site execution (single-GPU jobs, federated and asynchronous training, compressed and sharded inference, and pure forward-pass inference), and it explicitly excludes fully synchronous data-parallel training of frontier-scale models on tightly coupled accelerators. The exclusion is deliberate: the framework does not promise what it cannot deliver, and the workloads that fall outside its scope continue to motivate the parallel HVDC build-out that the nomadic paradigm complements rather than replaces. The eligibility test is a binary admission criterion at this stage of the work; relaxing it into a continuous metric is one of the open challenges identified at the end of the paper.

The rest of the paper develops the framework along this trajectory. Section II sets up the moving-bits-versus-moving-electrons framing and reviews the feasibility-domain model that constitutes the spatial degenerate case of the framework. Section III introduces the nomadic envelope, the three levers and their costs, the storage–compute equivalence principle, the unifying cost functional, and the eligibility test, and it recovers the prior work as the one-dimensional projection of the richer object. Section IV develops the multi-scale green infrastructure design that the envelope analysis requires, distinguishing nano-datacentres on low voltage from micro-datacentres on medium and high voltage, and assembling the silicon–lithium isopreference frontier as the quantitative interface between compute build-out and battery build-out at the system scale. Section V describes the technological stack that turns the framework into a deployable artefact, with a programmable data plane on SRv6 and IOAM substrates, a unified measurement plane that correlates network, compute, and energy signals at sub-second timescales, a multi-lever orchestrator that arbitrates among the four moves available at every decision step, and a verifiable green-compute certification format designed for standardisation. Section VI presents the Italian federated testbed on which the framework is being empirically validated, anchored on the PrinceLab cyber-physical platform at the Politecnico di Bari and on the GARR national research backbone, with a Llama-class federated continual learning workload as the primary load and an AI-RAN node as the second load at the tightest edge of the eligibility envelope. Section VII lists the open challenges that remain inside the framework, with a brief pointer to the separate companion document that articulates concrete tracks for European collaboration. Section VIII concludes.

II. MOVING BITS, NOT ELECTRONS

The European default response to the asymmetry sketched in Section I is the construction of transcontinental High Voltage Direct Current (HVDC) corridors [3]: SuedLink in Germany, the Tyrrhenian Link in Italy, the Mediterranean ELMED interconnector, the eastern bootstraps in the United Kingdom, collectively tens of billions of euros committed to moving electrons across the continent so that they may reach the compute that needs them. The dual question, *why not move the computation to where the electrons are, rather than the other way around?*, is the central premise of the present paper, and is the subject of a thin but growing body of work that we now review.

The premise is not novel as a slogan; variants of “follow-the-sun” geographically distributed computing have been articulated for more than a decade in the cloud and AI research communities [5]–[7] and have been deployed at hyperscaler scale as Carbon-Intelligent Compute Management [8], while energy-aware orchestration for AI training within a single facility has been recently revisited [9], [10]. A complementary independent line of work has formalised geographic load shifting between datacentres and the German energy grid as a residual-load optimisation problem over the four German TSO zones, using real generation data from the SMARD platform

of the Bundesnetzagentur [11]; that line converges on the same two axes (data versus energy, time versus geography) that the present paper formalises through the nomadic envelope. What has been missing is a *quantitative* answer to the underlying question that determines whether the slogan can become an operational paradigm: under which conditions can a real AI workload be physically moved across sites, fast enough to follow a renewable surplus window, and with energy and time costs that justify the move?

A first quantitative answer to this question was provided in our prior work [4]. Starting from realistic GPU checkpoint and migration technology [12]–[15] and from operational curtailment patterns observed on the California Independent System Operator and analogous European grids, the work formalises a *feasibility-domain model* for migratory single-GPU AI workloads. The model decouples three coupled factors that govern feasibility: the size of the migratable state of the workload S , the inter-site network bandwidth B , and the duration T_{energy} of the renewable-surplus window at the destination. A workload migration is *time-feasible* when the disruption introduced by the transfer fits within a fraction α of the available window:

$$T_{\text{transfer}} + T_{\text{load}} + T_{\text{downtime}} < \alpha T_{\text{energy}},$$

with $T_{\text{transfer}} = S/B$ dominating the other components for realistic checkpoint sizes. A complementary energy-breakeven condition checks that the energetic cost of moving the workload is repaid by the renewable-energy benefit at the destination.

Two findings from [4] carry into the rest of this paper. First, the energy-breakeven time is in the order of minutes, while typical renewable-surplus windows last hours; *time, not energy, is the dominant constraint* shaping feasibility. Second, feasibility in the (S, B) space partitions migratable workloads into three regimes (Class A, fully feasible; Class B, conditional; Class C, infeasible) that align cleanly with the typical checkpoint sizes of contemporary single-GPU AI jobs. The result is that for a large and operationally significant class of workloads, “moving bits instead of electrons” is not a future scenario but a tractable engineering problem, with a feasibility model that says *when* the move is worth making and *how* the orchestration should be designed around the resulting constraints.

The feasibility-domain model establishes the first leg of the GridAI paradigm: *a single AI workload can be relocated, in space, to follow renewable energy availability, under explicit and verifiable feasibility conditions*. The model is also explicit in its scope: it addresses single-GPU jobs with self-contained checkpoints, it treats migration as a one-shot decision rather than a continuous control problem, and it does not directly model two additional degrees of freedom that prove essential to a fully operational renewable-native AI paradigm. These two missing dimensions are the *temporal* flexibility of the workload (when must the result actually be available?) and the *storage-based* flexibility of the supporting infrastructure (how much energy can be buffered locally to extend or smooth the

renewable window?). The remainder of this paper develops the framework that adds these dimensions, and shows that doing so changes the design space of the renewable-native AI infrastructure in qualitative, not merely quantitative, ways.

III. ADDING THE TIME AXIS: THE NOMADIC AI COMPUTE ENVELOPE

A. From feasibility to envelope

The feasibility-domain model of [4] answers a binary question for a single workload at a single instant: *given a renewable-surplus window starting now at a candidate destination, can this workload be physically moved there in time?* The answer depends on a triple $(S, B, T_{\text{energy}})$, with the workload state size and the inter-site bandwidth on one side and the duration of the energy window on the other. This is a one-dimensional projection of a richer object. The model assumes that the workload *must run now or has just become ready to run*, and that the only flexibility available to the orchestrator is *where* to put it; it does not represent the additional flexibility that emerges once the orchestrator is also free to choose *when* the workload runs and *how much* energy is buffered locally to extend the renewable window.

A second axis is needed because real AI workloads differ by several orders of magnitude in how much delay they tolerate. The same Llama-class model serves a live conversational session that must respond within seconds, an interactive coding assistant that tolerates tens of seconds, an overnight fine-tuning job on a private codebase that tolerates a working night, and a periodic continual-learning sweep that tolerates several days between snapshots. At the level of the GPU the four workloads are indistinguishable; at the level of the orchestrator they live in completely different operating regimes. We call *deferrability* the application-level property that captures this distance between the moment a workload becomes ready and the latest moment at which its result is still useful, and we treat deferrability as a first-class input to the orchestration problem, on the same footing as the workload state size and the inter-site bandwidth.

The *nomadic envelope* of a workload is the space-time surface within which the workload can be executed while preserving its quality-of-service contract. Its two physical dimensions are the *spatial mobility* (the joule and time cost of relocating the workload between candidate sites, dominated for single-GPU jobs by the state-transfer term $T_{\text{transfer}} = S/B$) and the *temporal mobility* (the deferrability of the workload, measured in seconds for interactive services and in days for periodic batch jobs). The envelope is invisible to current AI orchestrators, which treat every workload as if it had zero deferrability and as if it were nailed to the site where it was admitted; making it explicit, measurable, and controllable is the central contribution of the GridAI framework.

B. Three levers, three explicit costs

Once both dimensions of the envelope are exposed, every orchestration choice becomes a selection among three physically distinct levers on the *bit* $\leftrightarrow$ *electron* $\leftrightarrow$ *time hinge*. Each lever

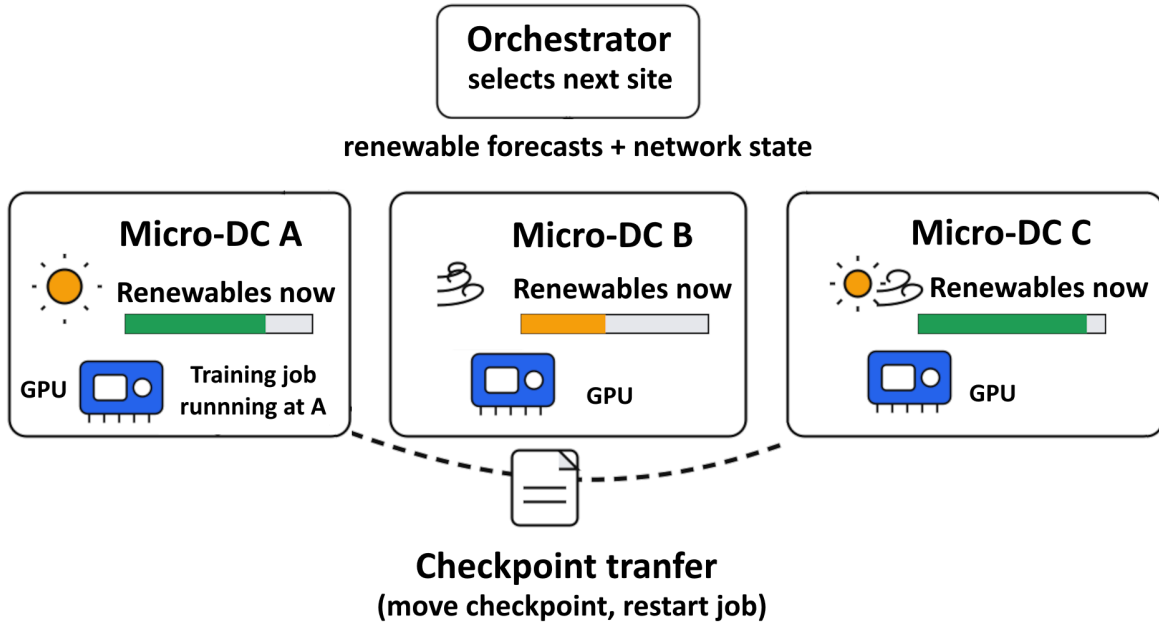


Fig. 1: Nomadic AI compute in operation. An orchestrator selects the next site for a training job from a fleet of energy-proximate micro-datacentres, on the basis of local renewable forecasts and inter-site network state. When the renewable surplus at site A is depleted, the job is checkpointed and restarted at site B or C where renewables are currently available; the spatial-mobility cost is paid in WAN bandwidth and stop-the-world time rather than in HVDC transport. The figure depicts the one-dimensional case of the framework, which Section III extends to the two-dimensional nomadic envelope by adding the temporal mobility of the workload.

has an explicit, quantifiable marginal cost; the orchestrator does not eliminate cost, it selects which cost to pay.

Moving bits relocates the workload to a site where renewable energy is currently abundant. The cost is communication: the energy and time of transferring the migratable state across the wide-area network, plus the synchronisation bandwidth that the workload will consume from the destination once it is running. For a single-GPU job, the dominant term is the one-shot transfer $T_{\text{transfer}} = S/B$ already captured by the feasibility-domain model. For a federated or sharded job, an additional steady-state term accounts for inter-site synchronisation traffic, whose cost can be reduced through gradient compression, sparse synchronisation, or controlled asynchrony, all of which trade convergence quality for communication budget. The benefit avoided is the high-voltage transport of electrons, with its resistive losses, conversion stages, infrastructure amortisation, and permitting timelines.

Moving electrons relocates the energy instead of the workload, sending power from a generation site to the computation site through the transmission grid, typically via High Voltage Direct Current (HVDC) corridors at continental scale [3]. The cost is the full HVDC chain: resistive losses on the line, AC/DC conversion losses at the converter stations, capital expenditure amortised across the line’s life, and the multi-year permitting and construction horizon. The benefit avoided is the relocation of compute, which is convenient when the workload genuinely cannot be moved (FullSync data-parallel training of

frontier models on tightly coupled accelerators is the canonical example) or when the geographical asymmetry between generation and consumption is structural and persistent.

Moving in time defers the workload until the renewable surplus arrives *in situ*, instead of chasing it in space. This lever has two physically distinct sub-modes that share the same algebraic effect on the carbon balance. The first sub-mode is *battery storage*: the surplus is buffered locally and released later to drive the workload. The cost is the storage chain: capital expenditure of the battery, round-trip efficiency losses (typically ten to fifteen percent for lithium-ion, larger for longer-duration chemistries), and calendar plus cycle degradation that erodes the asset over its life. The second sub-mode is *compute acceleration on the deferrable backlog*: given a surplus now and a queue of deferrable workloads that would have run later anyway, the surplus is consumed by accelerating those workloads, freeing the corresponding energy budget at their original scheduled time. The cost is the spent deferrability of the accelerated workloads, plus any over-provisioning of compute capacity required to absorb the surplus when it arrives.

C. The storage–compute equivalence principle

The two sub-modes of moving in time deserve to be put on the same footing, because their effect on the system’s overall carbon balance is, modulo specific costs, identical. Given a renewable surplus of E joules at time t , the orchestrator may either charge a battery by E (less the round-trip loss) and

The nomadic envelope of an AI workload

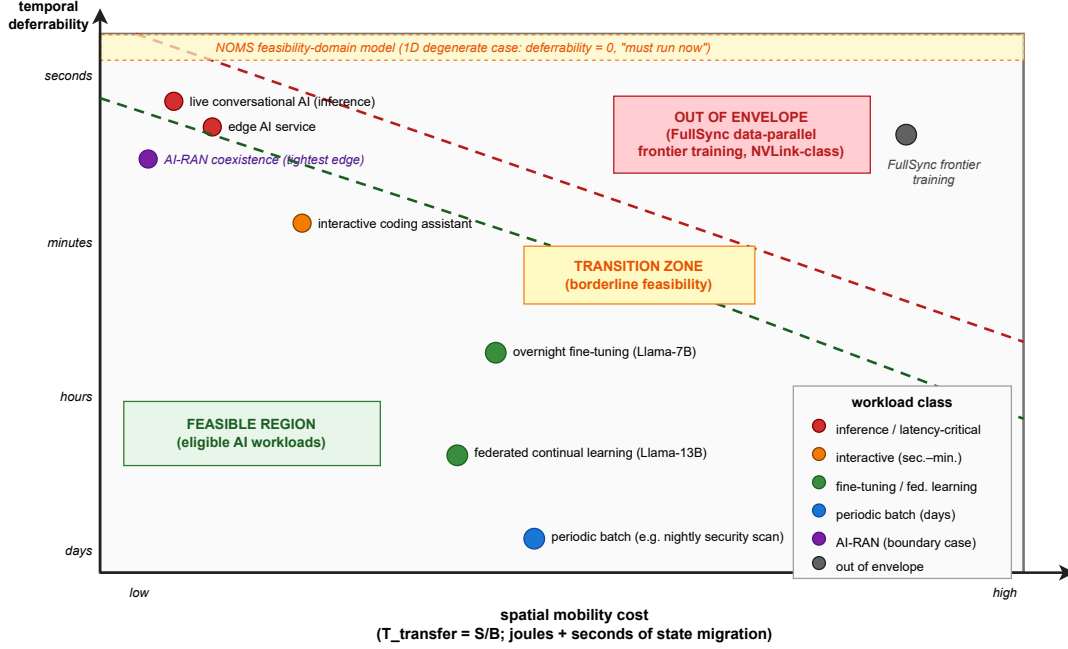


Fig. 2: The nomadic envelope of an AI workload: horizontal axis is the spatial mobility cost ($T_{\text{transfer}} = S/B$), vertical axis is the temporal deferrability of the workload, plotted with zero deferrability at the top (where the NOMS 2026 feasibility-domain model [4] is recovered as the 1D degenerate case) and high deferrability at the bottom. Two parallel oblique boundaries delimit a feasible region of eligible workloads, a transition zone of borderline feasibility, and an out-of-envelope region for FullSync data-parallel frontier training on NVLink-class fabrics that the framework explicitly excludes. Representative workloads are placed inside the space: from latency-critical inference, edge AI and AI-RAN coexistence at the top, through interactive coding in the upper transition, to overnight fine-tuning, federated continual learning and periodic batch progressively further down; FullSync frontier training sits at the top-right as the canonical out-of-envelope case.

release the energy at time $t' > t$ to run a workload that was already scheduled at t' , or it may consume E at time t to accelerate a deferrable workload pulled forward from its later scheduled time $t' > t$. In both cases, the renewable energy that would otherwise have been curtailed is converted into useful compute; in both cases, the energy budget at time t' is freed. The difference between the two modes lies entirely in their cost structure: battery storage pays a capital cost in lithium and a round-trip efficiency loss, while compute acceleration pays a capital cost in silicon and a deferrability cost on the accelerated workload.

We refer to this observation as the *storage–compute equivalence principle*, and we make it operational by defining the system’s *aggregate flexibility* as

$$F(t) = S(t) + D(t),$$

where $S(t)$ is the state of charge of the physical storage and $D(t)$ is the *equivalent state of charge* contributed by the deferrable backlog, measured as the renewable energy that could be absorbed by accelerating the workloads currently waiting in the queue. The orchestrator’s job is to keep $F(t)$ aligned with the expected curtailment trajectory, with the

freedom to invest in either the S component (lithium) or the D component (silicon plus deferrability buffer) depending on the marginal cost of each. This degree of freedom is the second contribution of the GridAI framework: **compute is virtual power storage for deferrable AI workloads**, and a portion of what is today called the “national battery” can be built in silicon rather than in lithium, with the partition between the two as an explicit design variable rather than an implicit assumption.

D. A unifying cost functional

The three levers and the equivalence principle can be assembled into a single cost functional that the orchestrator minimises along the lifetime of each workload. Let w index a workload, p index a placement decision (which site at which time), and ρ index a regime of execution (synchronous, asynchronous, compressed, sharded, and so on). We write the per-workload cost as

$$C(p, \rho, w) = C_{\text{bit}}(p, \rho, w) + C_{\text{electron}}(p, w) + C_{\text{time}}(p, w),$$

where the three terms count, respectively, the communication cost (joules and seconds of migration plus synchronisation

The bit ↔ electron ↔ time hinge: three levers, three explicit costs

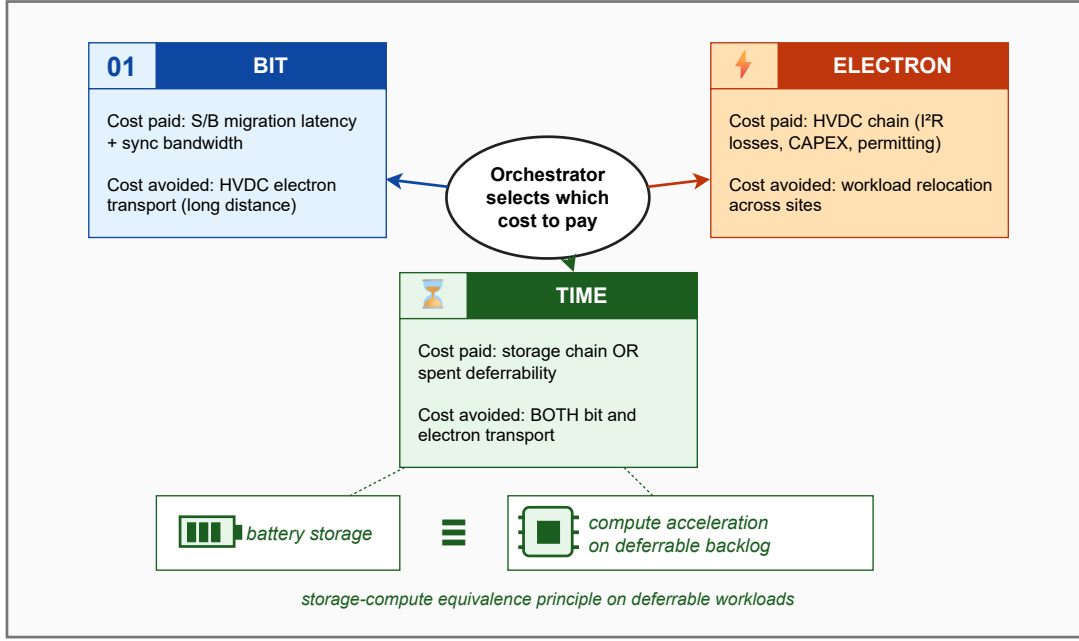


Fig. 3: The bit ↔ electron ↔ time hinge of nomadic AI compute: every orchestration choice selects one of three physically distinct levers, each with an explicit marginal cost. Moving bits pays the cost of state migration latency and synchronisation bandwidth, avoiding high-voltage electron transport. Moving electrons pays the cost of the HVDC chain (resistive losses, conversion stages, CAPEX, permitting horizons), avoiding workload relocation. Moving in time pays either the storage chain (battery CAPEX, round-trip losses, calendar and cycle degradation) or the spent deferrability of the workload, and avoids both bit and electron transport. The time lever has two sub-modes, battery storage and compute acceleration on the deferrable backlog, that are interchangeable on the system’s carbon balance modulo their specific costs (storage–compute equivalence principle, Section III-C).

under regime ρ), the electron-transport cost (HVDC chain attributable to the energy delivered to site p), and the time-flexibility cost (storage round-trip loss or spent deferrability, depending on which sub-mode is used at p). The orchestrator selects (p, ρ) for each workload so as to minimise the aggregate $\sum_w C(p_w, \rho_w, w)$, subject to the workload’s quality-of-service contract and to the system-level flexibility constraint that $F(t)$ remains within feasible bounds.

This formulation is deliberately decoupled from any specific optimisation technology. At the operational scale, C can be approached with model-predictive control or chance-constrained programming; at the longer timescale of capacity planning, it can be approached with mixed-integer programming over scenario ensembles; the framework is agnostic to the choice. What matters is that the three levers are *co-priced* along an explicit cost functional: today they are priced separately by different actors (grid operators price electrons, cloud providers price compute, battery vendors price storage) and the absence of a common metric is one of the structural reasons why the European policy debate has so far concentrated on a single lever (move electrons through new HVDC corridors), while

the comparison with complementary axes has not yet been fully formalised.

E. Eligibility, not envelope size

Not every AI workload is amenable to nomadic execution. The framework applies to workloads whose algorithmic regime is compatible with the bandwidth, latency, and synchronisation budget of inter-site execution: single-GPU jobs that fit entirely on one accelerator at the destination, federated training and inference that synchronises rarely and in compressed form, asynchronous schemes that tolerate stale gradients, sharded inference where the model is partitioned across sites with traffic shaped by the placement, and any pure inference workload that runs forward only. FullSync data-parallel training of frontier-scale models, where every micro-batch requires a tight AllReduce across thousands of accelerators inside a single high-bandwidth fabric, is out of scope by construction: a 400 Gbps NVLink domain is not reproducible across a continental WAN at any level of orchestration sophistication, and the framework does not promise what physics does not allow.

The eligibility test is a binary admission criterion, not a measure of the size of the envelope. A workload either belongs to the eligible class (in which case its nomadic envelope is well-defined and the cost functional is meaningful) or it does not (in which case it must be served by a co-located large facility and contributes to the demand that motivates the HVDC corridors). Empirically, the eligible class covers a large and operationally significant fraction of contemporary AI workloads: most inference traffic, most fine-tuning, most continual learning, most federated training, and most edge AI services fall into it, and the share is increasing as model architectures move toward sparsity and mixture of experts [16] and toward post-training adaptation [17] rather than ever-larger dense pretraining runs. The bet of the GridAI framework is that the eligible class is large enough to make the renewable-native paradigm relevant at system scale, even while the ineligible class continues to drive a parallel and complementary infrastructure conversation.

The tightest edge of the eligibility test is the AI-RAN node, where general-purpose AI workloads must coexist on the same accelerator with a millisecond-scale radio service whose envelope is by construction zero; any AI workload running on that hardware inherits a nomadic envelope shrunk by the radio’s residual capacity, and serves as the tightest stress test of the framework. We return to AI-RAN as the second application load of the federated testbed in Section VI, where it is developed both as a deployment site and as an experimental validation of the envelope-shrinkage predictions.

F. NOMS as the one-dimensional degenerate case

The feasibility-domain model of [4] is recovered exactly when the nomadic envelope is collapsed onto its spatial axis. Setting deferrability to zero removes the time lever; setting the local storage to zero removes the second sub-mode of the time lever; restricting the workload class to single-GPU jobs with self-contained checkpoints removes the regime variable ρ from the cost functional. What remains is a binary feasibility test on a single workload at the instant the renewable window opens, governed by the triple $(S, B, T_{\text{energy}})$ and parameterised by the disruption tolerance α . This is the NOMS model, verbatim. The richer framework therefore does not invalidate the prior work; it embeds it as the limiting case obtained when two of the three levers are switched off, and it inherits the prior work’s quantitative results (energy-breakeven in minutes, time-not-energy as the dominant constraint, Class A/B/C partition of single-GPU checkpoints) as boundary conditions that any orchestrator must respect on the spatial axis.

The relationship between the two models is therefore one of containment, not competition. The feasibility-domain model answers the question *can this workload be moved in space?*; the nomadic envelope answers the broader question *within which space-time region can this workload be executed, and which combination of levers minimises the cost of staying inside that region?* Each level of the analysis is meaningful on its own terms: a system that solves only the feasibility question already extracts value from spatial flexibility, and a system

that solves the full envelope problem extracts additional value from temporal flexibility and from the storage–compute trade-off. The two models are designed to be deployed together, with the feasibility check serving as a fast inner loop inside the broader envelope optimisation.

IV. MULTI-SCALE GREEN INFRASTRUCTURE DESIGN

A. Why four scales, and not one

The nomadic envelope is a property of the workload, but the physical sites among which the workload migrates have properties of their own. A renewable-native AI infrastructure cannot be a single class of facility scaled up or down; the distribution of renewable generation across the territory is intrinsically multi-scale, ranging from kilowatt-scale rooftop photovoltaic to gigawatt-scale wind farms, and the compute infrastructure that follows it must mirror that distribution if the spatial-mobility lever of the envelope is to be operationally meaningful. We propose a four-scale taxonomy of green compute sites, summarised in Table I and grouped into two categories on the basis of the electrical-system connection domain. The first category, *nano-datacentre*, comprises the two scales connected to the low-voltage distribution network (230/400 V) and operating entirely in the Distribution System Operator (DSO) domain: a single household with rooftop photovoltaic, residential storage and a small AI accelerator (Home-DC), and a residential or small commercial building with shared photovoltaic, building-scale storage and a rack of accelerators (Building-DC); both can aggregate their flexibility through the DSO local flexibility market. The second category, *micro-datacentre*, comprises the two scales connected to the medium-voltage network (15–20 kV in Italy) and, at the largest end, to the high-voltage transmission network: a facility colocated with a medium-voltage substation serving a neighbourhood-scale catchment (Neighbourhood-DC), and a facility colocated with a large renewable plant or with a structural grid bottleneck (Hot-spot-DC), where the 10 MW connection threshold separates the DSO and TSO domains. Figure 4 provides the visual taxonomy and Table I reports the corresponding numerical envelope.

The naming convention (nano on low voltage, micro on medium voltage and above) is internal to the GridAI framework and is motivated by the SI prefix progression, by the clean mapping onto the two voltage levels of the Italian electrical system, and by the academic precedent of the *NaDa* nano-datacentre line of work in the European FP7 programme [18].

The four scales differ not only in size but in the structure of the trade-off between local consumption and inter-site migration. At the Home-DC scale, the local renewable surplus is small and bursty, the local deferrable backlog is limited to a single user, and the relevant flexibility comes almost entirely from the spatial lever: workloads either run locally during a surplus window or migrate elsewhere. At the Hot-spot-DC scale, the local surplus is large and persistent, the catchment of deferrable workloads is system-wide, and the storage–compute trade-off dominates: most of the value is captured by deciding

GridAI four-scale infrastructure

Two categories on the connection domain of the Italian electrical system

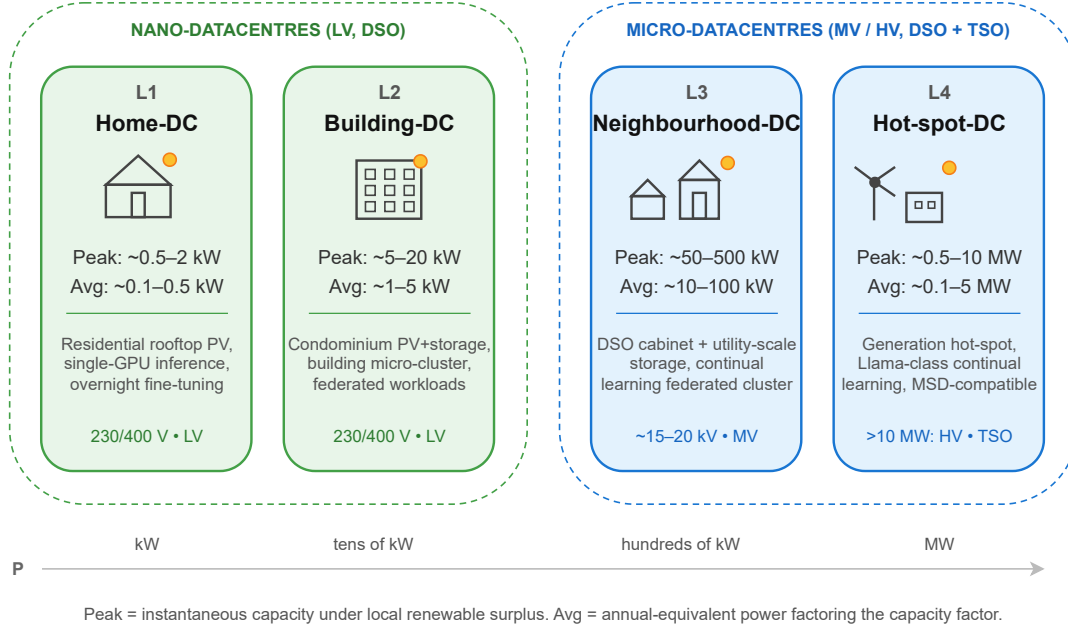


Fig. 4: The four-scale GridAI infrastructure. Nano-datacentres (L1 Home-DC, L2 Building-DC) operate on low voltage (LV, 230/400 V) in the DSO domain; micro-datacentres (L3 Neighbourhood-DC, L4 Hot-spot-DC) operate on medium voltage (MV, ~15–20 kV) in the DSO domain, with extension to high voltage (HV) and prospective TSO domain at the largest scale once installed capacity exceeds ~10 MW. Peak figures denote instantaneous usable capacity under local renewable surplus; annual-equivalent figures factor in the capacity factor of the local renewable profile.

TABLE I: The four scales of green compute sites in the GridAI taxonomy.

Scale	Voltage domain	Peak power	Avg power	Grid actor
<i>Nano-datacentre</i> (low-voltage, DSO domain)				
Home-DC	LV 230/400 V	0.5–2 kW	0.1–0.5 kW	DSO
Building-DC	LV 230/400 V	5–20 kW	1–5 kW	DSO
<i>Micro-datacentre</i> (medium- and high-voltage)				
Neighbourhood-DC	MV 15–20 kV	50–500 kW	10–100 kW	DSO
Hot-spot-DC	MV 15–20 kV / HV (≥ 10 MW)	0.5–10 MW	0.1–5 MW	DSO (< 10 MW); TSO (≥ 10 MW)

how much of the local surplus is buffered locally (in a co-located battery or in compute over-provisioning) and how much is dispatched as a flexibility service to the TSO. The intermediate scales sit on a continuum between these two extremes, and the four-scale design ensures that the same nomadic envelope framework can be instantiated coherently at every point of the continuum.

B. Hardware characterisation and 2026–2030 roadmap

The hardware substrate of the nomadic envelope is in rapid evolution along three axes that the framework must track and exploit. On the accelerator axis, the Blackwell generation (B100, B200, GB200 NVL72 systems) is the current frontier and is being deployed at scale through 2026 and 2027; the post-Blackwell generations announced for 2027 and beyond

will move further toward higher local memory capacity, lower per-token energy, and tighter integration with co-packaged photonics, all of which directly affect the migratable state size S and the per-token energy entries of the cost functional. On the data-path axis, the rise of DPU and IPU platforms (Blue-Field, IPU-class accelerators from the hyperscaler ecosystem) introduces a programmable network and storage layer adjacent to the GPU, and creates the possibility of offloading both the migration mechanics and the per-flow energy accounting to a dedicated programmable substrate; we return to this in Section V. On the storage axis, the lithium iron phosphate (LFP) chemistry has displaced nickel-manganese-cobalt for stationary applications on cost and safety grounds, sodium-ion chemistries are entering the market for cost-sensitive stationary

deployments, and longer-duration chemistries such as iron-air and vanadium-redox flow are reaching commercial pilot scale; each chemistry has a distinct cost, round-trip efficiency, and degradation profile, and the storage–compute equivalence principle of Section III requires the framework to be explicit about which chemistry is assumed at which site.

The roadmap also has a software dimension that conditions the hardware analysis. The relevant question at each scale is not just *how much compute do we install?* but *how is that compute virtualised, partitioned, and exposed to the orchestrator?*. Multi-instance GPU partitioning, time-sliced sharing, and confidential computing primitives all change the effective granularity at which a workload can be placed on a site, and therefore the effective dimensionality of the placement variable p in the cost functional. A workload that requires a full GPU sees a different envelope than a workload that fits inside a one-seventh GPU slice; the framework must surface this granularity rather than averaging over it.

C. Cloud orchestration and federation across scales

The four scales must be operated as a single federated computing fabric, not as four disjoint deployments. The orchestration plane is responsible for admitting workloads at the edge they reach first (typically a Home-DC, a Building-DC, or a Neighbourhood-DC), classifying them along the eligibility test of Section III-E, computing their nomadic envelope from their declared QoS contract, and then routing them across the scales according to the minimisation of the cost functional. The natural substrate for this layer is a container-based orchestration system extended with energy-aware scheduling primitives, a service mesh that exposes per-site energy carbon-intensity and renewable-availability metrics as first-class scheduling attributes, and a CPU and GPU virtualisation layer that supports live migration of stateful AI workloads across heterogeneous hardware.

Two technical observations shape the design of this layer. First, the migration primitive at the heart of the spatial lever is not a generic “cold restart from checkpoint” operation; recent work on transparent GPU checkpointing [12], [13], hot migration of inference services [14], [15], and disaggregated storage of model state has shown that single-GPU migration with sub-second downtime is achievable for a wide class of workloads when the system stack is co-designed across the GPU runtime, the network, and the orchestrator. Second, the federation must coexist with the existing commercial cloud rather than replacing it: most workloads will continue to enter the system through commercial cloud edges and will be routed into the nomadic infrastructure only when their envelope makes it cost-effective; the federation therefore has to expose itself as a standard set of compute endpoints that the upstream cloud orchestrator can address, and the energy-aware policy has to be enforceable at the boundary rather than only inside the federation.

D. Coupling with the national power grid

The four-scale design only delivers system-level value if the compute infrastructure is coupled, in a controllable way, to the power grid that hosts it. The coupling has two distinct flavours, corresponding to the two electrical domains. In the DSO domain (Home-DC, Building-DC, Neighbourhood-DC, and the smaller end of Hot-spot-DC), the natural integration mechanism is the local flexibility market that the European Clean Energy Package has mandated and that ARERA is operationalising in Italy through the regulatory framework on TSO/DSO coordination and DER aggregation. GridAI compute sites act, in this domain, as Distributed Energy Resources (DER) whose flexibility (the time and amount of energy they can shift, given their backlog of deferrable workloads and their local storage state of charge) is exposed to the DSO through standard aggregation interfaces. In the TSO domain (the upper end of Hot-spot-DC, above the 10 MW threshold), the same flexibility can in principle participate in the ancillary-services and balancing markets operated by Terna, subject to the prequalification and metering requirements of those markets; we treat this as a prospective extension rather than as a baseline assumption, because the regulatory and technical maturity of compute-as-flexibility on the TSO side is still in development.

A consistent grid-coupling analysis requires the framework to simulate, at each scale, the closed loop between the local renewable generation profile, the local compute consumption profile, the local storage trajectory, and the dispatch signals coming from the DSO or TSO operator. This simulator is one of the central deliverables of the GridAI methodology and is designed to be open-source and reproducible. The simulator takes as inputs (i) historical or synthetic renewable generation traces at the relevant scale, (ii) historical or synthetic AI workload traces with declared deferrability profiles, (iii) the hardware roadmap of the candidate site, (iv) the storage chemistry and sizing, and (v) the regulatory parameters of the DSO or TSO domain. It produces as outputs the trajectory of $F(t)$, the realised value of the cost functional, the share of workloads served by each lever, and the avoided cost of the corresponding HVDC alternative.

E. The silicon–lithium isopreference frontier

The combination of the four-scale infrastructure design, the storage–compute equivalence principle, and the simulator described above makes possible a quantitative answer to a question that today is asked only in policy terms. The question is: *how much of the national flexibility envelope should be built in lithium, and how much in silicon?* Each site can be sized along a continuum that ranges from a pure-storage configuration (large battery, minimum compute over-provisioning, deferrable workloads served at their nominal schedule) to a pure-compute configuration (minimum storage, large compute over-provisioning, deferrable workloads accelerated whenever a surplus arrives). Between the two extremes lies a continuous frontier along which the system delivers the same flexibility service to the grid at different splits between lithium capital

expenditure and silicon capital expenditure; we call this the *silicon–lithium isopreference frontier* of the site.

The frontier moves over time as the costs of the two technologies evolve. Battery costs have fallen by an order of magnitude over the past decade and are expected to continue declining, although at a slower pace; accelerator costs have not fallen at the same rate but have improved sharply on the per-token energy axis, which is the axis that matters for the silicon side of the trade-off. The framework does not predict the future evolution of the two cost curves; it makes the trade-off explicit and computable, so that policy decisions on grid-scale storage procurement and on the public co-funding of compute infrastructure can be taken with a shared metric rather than with two parallel and uncoordinated processes. This is, in our view, the most consequential policy contribution of the multi-scale infrastructure analysis: it gives the regulator, the TSO, and the cloud providers a common axis along which to negotiate the build-out of the renewable-native flexibility envelope.

F. Avoided cost of HVDC alternatives

The macro-economic value of the four-scale design must finally be compared to the alternative against which it is positioned: the HVDC build-out that is currently the European default. The total capital expenditure committed across the announced European HVDC corridors is in the order of tens of billions of euros, with multi-decade amortisation horizons and the permitting, right-of-way and construction lead times typical of continental-scale transmission infrastructure. The avoided-cost analysis associated with the multi-scale infrastructure is not a claim that HVDC corridors are unnecessary; the eligibility test of Section III-E makes clear that a substantial class of AI workloads cannot be served by the nomadic paradigm, and the corresponding demand will continue to motivate continental electron transport. The share of curtailed renewable energy that GridAI can plausibly address is, by construction, a fraction of the total HVDC-targeted volume; equivalently, the share of AI workloads that can be moved away from centralised hyperscaler facilities into energy-proximate infrastructure is a fraction of the global AI compute mix. In both cases the fraction is small in relative terms but large in absolute terms, both in carbon-balance impact and in industrial-investment value, and large enough to justify a dedicated infrastructure alongside the HVDC build-out rather than as a competitor to it. What the analysis does is quantify the share of the projected continental energy transport that can be displaced, at lower system cost, by serving the eligible AI workload class through the four-scale nomadic infrastructure. A credible answer to this question, even an order-of-magnitude answer, would already be sufficient to inform the next decade of European infrastructure planning; producing such an answer, calibrated on real Italian grid and workload data, is one of the headline objectives of the GridAI framework.

V. TECHNOLOGICAL STACK: PROGRAMMABLE DATA PLANE, ORCHESTRATION, CERTIFICATION

A. A programmable data plane for nomadic compute

The cost functional of Section III-D requires, on the bit lever, an accurate and time-resolved measurement of the energy and time consumed by every relevant traffic flow. This measurement cannot be reconstructed after the fact from coarse aggregate counters; it must be produced *in the data plane*, at line rate, by the same forwarding elements that move the bytes. The technical substrate for this requirement is the programmable data plane that has emerged over the past decade around three converging families of technologies: source routing extensions of the IP layer (SRv6, SR-MPLS), in-band telemetry primitives (IOAM and its successors), and programmable network interface cards built around DPU and IPU platforms.

The GridAI data plane is designed to combine these three families into a single coherent substrate. SRv6 provides the steering primitive that lets the orchestrator pin a workload’s traffic onto a specific inter-site path chosen for its energy profile, rather than letting the path emerge from default interior-gateway routing decisions that are unaware of the carbon intensity along the way. IOAM provides the per-packet telemetry that exposes, at the egress of each programmable hop, the energy attributable to the flow, the queuing delay it has accrued, and the residual capacity along the path; this telemetry is consumed by the orchestrator as a closed-loop signal that closes the gap between the modelled cost functional and its realised value. The DPU and IPU layer provides the physical site at which both functions are anchored: a per-server programmable element, sitting on the path between the GPU and the network, that runs the SRv6 endpoint logic, the IOAM telemetry generation, and (where applicable) the live migration mechanics of the workload itself.

The data plane is not a single static implementation but an *adaptive* one. Three forwarding regimes coexist: the standard Linux kernel stack, an eBPF/XDP fast path that bypasses the heavier parts of the kernel for energy-sensitive flows, and a full kernel-bypass regime (DPDK or vendor-equivalent) for the most demanding inter-site synchronisation traffic. The orchestrator selects among the three regimes as a function of the workload’s energy profile and of the available energy budget at the site: when energy is plentiful, the standard stack is acceptable; when energy is constrained, the lighter-weight regimes are activated to reduce the per-packet energy cost; the selection is reversible at the flow level rather than being a static deployment choice.

B. A unified measurement plane

The data plane sketched above is part of a broader measurement infrastructure that crosses the boundary between the network and the compute. The cost functional aggregates terms that originate in three distinct subsystems: the network (per-flow energy and time), the compute (per-token or per-iteration energy at the GPU), and the energy plant

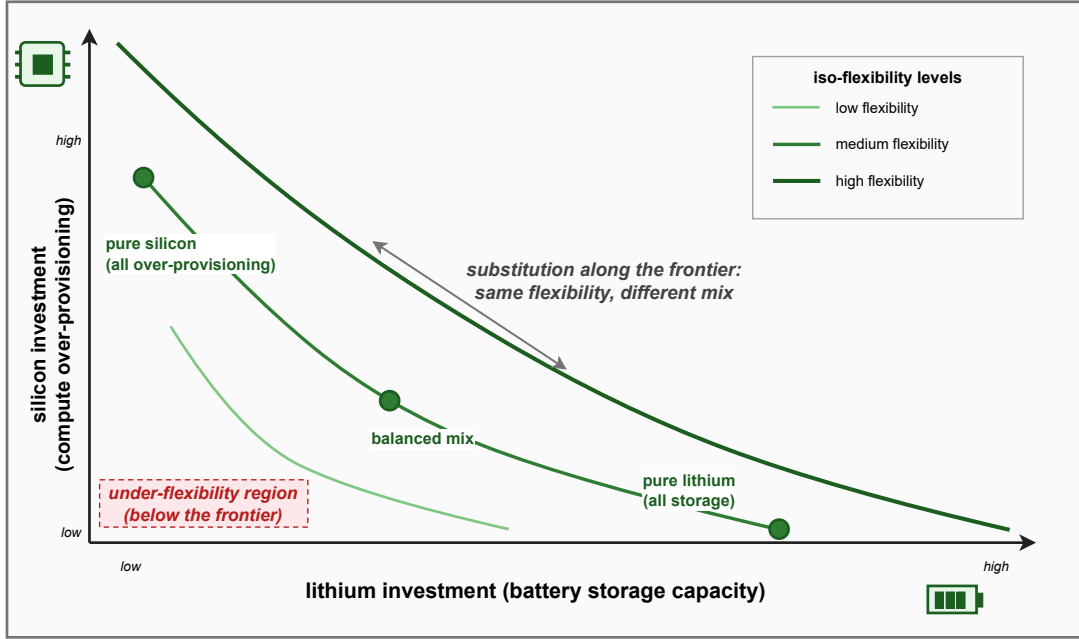


Fig. 5: The silicon-lithium isopreference frontier of a GridAI site. Each curve traces the locus of (lithium investment, silicon investment) pairs that deliver the same flexibility service to the grid, plotted for three indicative service levels (low, medium, high). The medium curve is annotated with two corner configurations and one balanced mix: pure silicon (all flexibility from compute over-provisioning on the deferrable backlog, minimum storage), balanced mix (a calibrated split between battery storage and compute capacity), and pure lithium (all flexibility from battery storage, minimum over-provisioning). Substitution along a curve preserves the flexibility level while trading one technology for the other (storage–compute equivalence principle, Section III-C); moving to a higher curve buys more flexibility at higher total CAPEX. The shaded region below the lowest frontier corresponds to under-flexibility configurations that do not meet the target service to the grid.

(instantaneous renewable generation, storage state of charge, grid carbon intensity). A coherent measurement plane must correlate the three subsystems at sub-second timescales and produce a unified time series in which the relationship between a workload’s compute output, its network footprint, and the energy it consumed is reconstructable end to end.

The technical primitives for this correlation are by now standard: eBPF-based probes at the kernel and userspace boundary for the compute side, XDP and TC hooks plus IOAM at the network side, and standardised industrial metering (e.g. via IEC 61850 and the related family of substation-automation standards) at the energy-plant side. What is missing today is the *glue layer* that publishes these signals against a common time base and a common per-workload identifier, so that the orchestrator and the certification subsystem (described below) can consume them as a single coherent stream rather than as three disjoint dashboards. Producing this glue layer, with an open and reproducible specification, is one of the central deliverables of the GridAI stack.

C. Traffic engineering for AI workloads

The eligible workload class identified in Section III-E is dominated by training and inference schemes that synchronise across sites through a small set of collective commu-

nication primitives: AllReduce for gradient synchronisation, scatter-gather for sharded inference, broadcast for parameter dissemination, and a handful of point-to-point patterns for federated and asynchronous regimes. These primitives are well-studied at the level of the high-performance computing intra-rack fabric, where they are typically realised on top of tightly coupled high-bandwidth interconnects. What is less well-studied is how the same primitives behave when they are mapped onto an inter-site WAN substrate, where the relevant constraints (bandwidth, latency, jitter, and now carbon intensity) are several orders of magnitude harsher than inside a single facility.

The data plane described above gives us the substrate for an explicit *data-plane mapping* of the collective primitives onto the inter-site fabric. A federated AllReduce, for example, can be decomposed into a tree of partial sums whose topology is chosen by the orchestrator as a function of the energy profile of the candidate paths, rather than being inherited from a generic best-effort routing decision; a sharded inference can be steered so that its scatter-gather traffic crosses the WAN only along paths whose carbon intensity is below a threshold set by the workload’s QoS contract. This is, in essence, a traffic-engineering problem with two extra dimensions (energy and

carbon) added to the classical (bandwidth, latency) optimisation space, and it is the natural sphere of contribution of the network research community to the GridAI framework.

D. A multi-lever control plane

The orchestration plane that consumes the measurement substrate and acts on the data plane is, in our view, the central technical artefact of the GridAI framework after the envelope itself. At every decision step, the orchestrator selects among four distinct moves for the bundle of energy and workloads currently available at a site: *consume now* (serve an urgent workload immediately, paying the corresponding cost on the electron and bit levers), *charge storage* (buffer the surplus locally, paying the storage round-trip and degradation cost), *accelerate the deferrable backlog* (pull a deferrable workload forward, spending its deferrability budget but absorbing the surplus into useful compute), or *curtail* (release the surplus, paying the opportunity cost of wasted renewable energy and the corresponding carbon footprint elsewhere on the grid). The four moves are co-priced along the cost functional and are not enforced separately by four different subsystems; this co-pricing is what differentiates the GridAI orchestrator from the current generation of carbon-aware schedulers, which typically expose only the spatial lever and treat storage and deferrability as exogenous parameters. Figure 6 sketches the resulting architecture: an Overall Orchestrator coordinates the Network and Infrastructure orchestrators, and a vertical Workload Mobility Manager handles the as-a-whole transfer of workloads across the energy-proximate sites of the federated physical infrastructure.

The orchestrator is structured as a multi-paradigm decision-making stack, in the sense that no single algorithmic family is suitable across all timescales of the problem. At the longest timescale (months to years, capacity planning), the relevant tools are mixed-integer programming over scenario ensembles of renewable generation and workload demand. At the intermediate timescale (hours to days, energy-aware admission and migration), chance-constrained programming and model-predictive control are the natural fit, because the decisions must hedge against uncertain renewable forecasts and uncertain workload arrivals. At the shortest timescale (seconds to minutes, runtime lever selection), deep reinforcement learning agents trained on the simulator of Section IV-D are the most promising avenue, because the decision rate is too fast for explicit optimisation and the state space is too large for tabular methods. The three layers must be co-designed rather than stacked naively: the long-horizon plan sets the parameters of the intermediate-horizon optimisation, which in turn shapes the reward function of the runtime agent.

The orchestrator is also *intent-based*, in the sense that the workload's QoS contract is expressed declaratively (deadline, eligible regime, accepted JCT inflation, carbon intensity target) rather than imperatively as a placement. The translation from intent to enforced plan is performed by the orchestrator and is auditable through the certification layer described below; this design choice aligns the GridAI orchestrator with the

broader trajectory of intent-based networking and intent-based service management, and is also what makes the agentic decomposition of the orchestrator feasible. A modern AI-centric orchestrator can decompose its task into a hierarchy of agents that negotiate among themselves (a placement agent, a storage agent, a backlog agent, a certification agent), each operating on a slice of the cost functional and converging on a joint plan that is then enacted on the data plane.

E. KPI/KVI value-driven evaluation

The orchestrator's performance must be evaluated on metrics that go beyond the classical Key Performance Indicators (KPIs) of cloud and network engineering. The accepted KPI set (latency, throughput, JCT, availability, error rate) remains necessary but is no longer sufficient: a renewable-native infrastructure must also surface Key Value Indicators (KVIs) that quantify the carbon, energy, and grid-flexibility footprint of the service. We therefore align the GridAI evaluation with the KPI/KVI duality that has been articulated in the 6G and beyond-5G community, and we extend it with two GridAI-specific indicators: joules per token (energy efficiency of the workload along its actual execution path) and grams of CO₂-equivalent per token (carbon efficiency of the same path, weighted by the carbon intensity of the energy that drove it). Both indicators are measured in line by the unified measurement plane and are published per workload, per site, and per time window.

The KVI framing is not a presentational choice; it directly conditions the orchestrator's reward function and the certification subsystem's audit criteria. A workload whose intent specifies a carbon-intensity cap below a given threshold is steered, by the multi-lever control plane, onto the combination of placement and regime that minimises the carbon component of the cost functional subject to the KPI cap on JCT inflation. The orchestrator therefore cannot be evaluated only on its KPI performance and must publish, for every served workload, the realised KVI trajectory and the conditions under which the cost functional was minimised.

F. Verifiable green compute certification

A renewable-native AI infrastructure is only credible to its users, its regulators, and the broader public if the energy and carbon claims it makes about itself are independently verifiable. The current state of practice on green-compute claims is dominated by self-reported aggregates published at the level of the facility or of the cloud region, often computed on annual averages and with no end-to-end traceability between a specific workload and the specific energy that drove it. This is not adequate for a paradigm that aims to expose the bit, electron, and time levers to a market and to a regulator.

The GridAI certification layer is designed to produce *verifiable evidence as a byproduct of the orchestrator's decisions*, rather than as an after-the-fact audit overlay. At every decision step, the orchestrator emits a structured record that ties the workload identifier, the chosen lever combination, the measured energy and carbon footprint, and the source of the un-

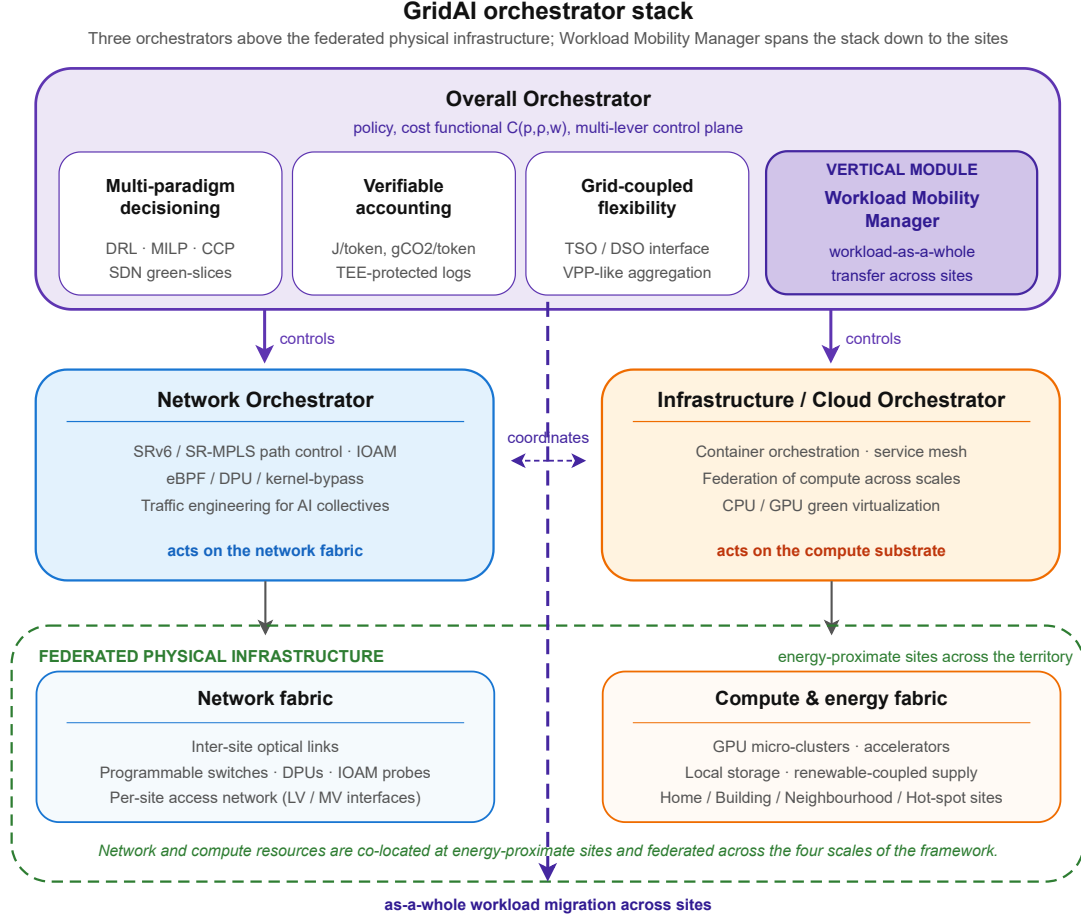


Fig. 6: The GridAI orchestrator stack. The Overall Orchestrator coordinates the Network Orchestrator (data plane) and the Infrastructure/Cloud Orchestrator (compute substrate); the Workload Mobility Manager is a vertical module of the Overall Orchestrator that handles the as-a-whole transfer of workloads across the energy-proximate sites of the federated physical infrastructure.

derlying renewable generation, into a single cryptographically-signed object. The signature is anchored on a cryptographic root-of-trust at the site (for instance a Trusted Execution Environment co-located with the GPU), so that the record cannot be retroactively rewritten by either the operator or an external party; the records are then committed to a tamper-evident, publicly verifiable log (for instance a Merkle-tree-based transparency log), so that any party (the workload owner, an external auditor, a regulator) can verify the chain of custody between a published green-compute claim and the underlying physical measurements. The format of the resulting *green compute certificate* is designed to be implementation-agnostic and to be standardisable; we intend to push the format toward an IETF Internet-Draft, with the Computing-Aware Traffic Steering (CATS) working group of the Routing area [19] as the primary target for the certificate format itself, and with the GREEN working group of the Operations and Management area [20] as the natural community venue for the broader

energy-efficient networking standardisation agenda in which GridAI inscribes itself; the aim is for the certificate to become a building block for cross-operator compatibility rather than a GridAI-internal artefact.

The certification layer also closes the loop with the grid-coupling described in Section IV-D. When a GridAI site exposes flexibility to a DSO market or, prospectively, to a TSO market, the dispatch and verification of the flexibility service requires a credible measurement of the energy actually shifted; the certificate produced by the orchestrator can be used directly as the metering object for these markets, eliminating the duplication between the cloud-side energy accounting and the grid-side flexibility settlement. This dual use of the certificate (audit for the workload owner, settlement for the grid operator) is one of the structural reasons why the certification layer is designed at the same level of priority as the data plane and the orchestrator, rather than being added as a peripheral concern.

VI. ITALIAN FEDERATED TESTBED AND END-TO-END DEMONSTRATION

A. Why a federated testbed, and why in Italy

The envelope analysis, the multi-scale infrastructure design, and the technological stack converge into a single empirical claim that must be testable end to end: *an eligible AI workload, with a declared deferrability profile, can be served by a federation of energy-proximate micro-clusters at a JCT inflation below a stated threshold and at a renewable share above a stated threshold, with a third party able to verify both numbers from the green-compute certificate emitted by the orchestrator.* The only way to substantiate this claim is to build the federation and to run the workload on it, with a measurement infrastructure thick enough to expose every term of the cost functional. A position paper that stops at the framework, without an empirical anchor, is too easy to dismiss as a methodological exercise; a deployment without the framework is too easy to dismiss as one of the many isolated proofs of concept that the carbon-aware computing community has accumulated over the past decade. The GridAI project is designed to deliver both at once, and to make the deployment the substantiation of the framework rather than a parallel artefact.

Italy is a particularly favourable setting for this empirical validation. The country's electrical system spans the full distribution-to-transmission stack relevant to the multi-scale design: a dense low-voltage distribution network with high residential photovoltaic penetration in the centre and north, a medium-voltage network with significant industrial and agricultural renewable installations across the south, and a high-voltage transmission network with several documented bottlenecks where renewable curtailment is structural. The 338 GWh of renewable energy curtailed in 2024 [2] is concentrated geographically and provides a natural test bed for the spatial-mobility lever: there are sites where the local renewable surplus is reliably available for measurable fractions of the year and where a credible workload migration target is therefore physically present. The Italian research backbone GARR, with its multi-Tbps fibre infrastructure and its presence at each of the consortium sites, provides the inter-site network substrate at the scale required for the federation. The combined availability of a regulated grid with mandated TSO/DSO coordination (under ARERA's framework), a high-curtailment renewable portfolio, a dense fibre backbone, and a consortium of universities each owning a credible candidate site is what makes Italy the right place to attempt the empirical validation.

B. Architecture of the federated testbed

The federated testbed is built on at least three consortium sites, each instantiating a different scale of the four-scale infrastructure design. The physical hub of the testbed is the *PrinceLab* cyber-physical platform at the Politecnico di Bari, which is the only facility in the consortium that integrates, on a single coherent infrastructure, distributed renewable generation (photovoltaic and storage assets at the building scale),

programmable networking equipment, and an AI micro-cluster sized at the upper end of the Building-DC scale. PrinceLab serves as the reference deployment for the smallest scale of the GridAI taxonomy and is the site where the storage-compute equivalence principle can be measured most directly, because the storage assets and the compute assets are physically co-located behind a single measurement boundary.

The other consortium sites instantiate larger scales of the taxonomy and complementary network roles. At the Neighbourhood-DC scale, candidate deployments at the Politecnico di Milano, the Politecnico di Torino, and the Federico II di Napoli host AI micro-clusters connected through the local DSO substation and integrated with site-level photovoltaic and battery storage; each of these sites contributes a different mix of renewable generation, workload mix, and local-flexibility regulatory context, which is the input needed to calibrate the envelope analysis across the heterogeneity of the Italian distribution network. At the largest end of the taxonomy, a Hot-spot-DC class deployment is contemplated as a stretch goal at a site adjacent to a structurally curtailing renewable plant, subject to the operational and regulatory feasibility of the connection. The Università di Pisa and the Roma Tor Vergata sites contribute the agentic orchestration substrate and the SRv6/IOAM-based network fabric respectively, both of which are deployed across all participating sites and provide the cross-site control and measurement planes.

All sites are interconnected through the GARR backbone, which provides multi-10 Gbps to 100 Gbps inter-site links and is, for our purposes, a representative substrate for the kind of inter-site bandwidth that is realistic in a continental research and education network or in a commercial inter-region cloud fabric. The choice of GARR as the inter-site fabric is deliberate: it ensures that the empirical results carry over to operationally meaningful settings rather than being tied to a dedicated experimental network engineered around the testbed.

C. The headline demonstration: federated continual learning of a Llama-class model

The primary workload of the federated testbed is a continual learning sweep of a Llama 3-class model in the 7 to 13 billion parameter range, executed in a federated regime across the at-least-three sites of the testbed. The choice of Llama-class is motivated by representativeness rather than by performance ambition: this scale of model captures the operational regime of contemporary single-GPU fine-tuning and continual learning, which is the largest single contributor to the eligible workload class identified in Section III-E. The training regime is qualified in detail because the framework does not promise what it cannot deliver: synchronisation is sparse (one global aggregation per several minutes of local training rather than per micro-batch), gradients are compressed (top- k sparsification together with low-precision quantisation along the lines that the federated learning community has converged on), and asynchrony is controlled (the global aggregation tolerates straggler sites within a bounded staleness budget). FullSync

Architecture of the GridAI federated testbed

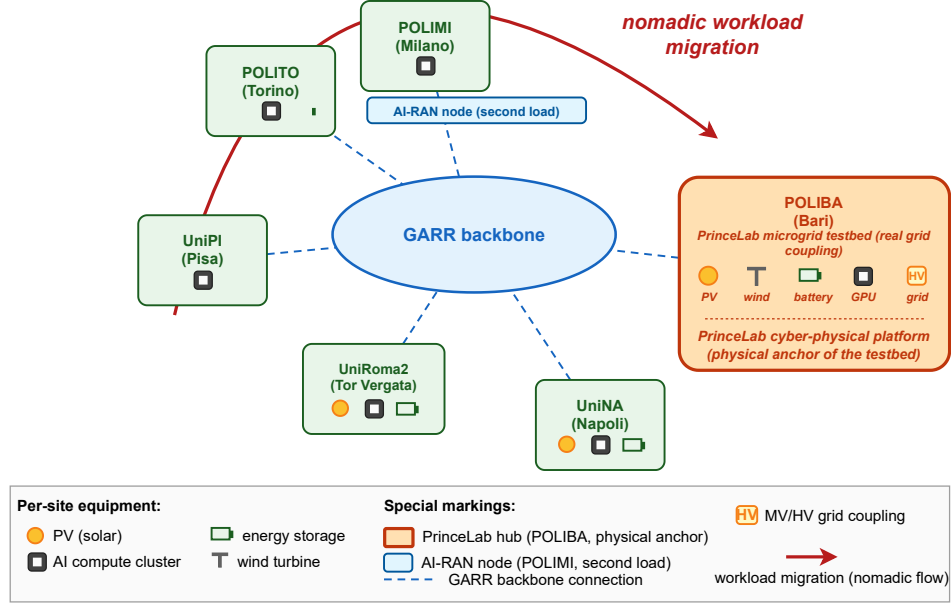


Fig. 7: Architecture of the GridAI federated testbed: six Italian consortium sites interconnected via the GARR national backbone (central blue oval). Three sites (POLITO, POLIMI, UniPI) host compute-only AI nodes; two sites (UniRoma2, UniNA) are energy-proximate, adding on-site photovoltaic and battery storage to the AI compute cluster; POLIBA hosts the PrinceLab cyber-physical platform (highlighted in orange), the physical anchor of the testbed, which adds wind generation and a real MV/HV grid coupling to the standard equipment set and is the only consortium site where the storage–compute equivalence principle can be measured behind a single metering boundary. POLIMI hosts the AI-RAN node as the second application load (Section VI-D). The red curved arrow illustrates a representative workload migration along the spatial-mobility lever of the nomadic envelope.

data-parallel training of the same model is, deliberately, out of scope: the framework would not survive a tight AllReduce across the GARR backbone, and we will not claim a result that we cannot honestly deliver.

The demonstration is structured around three quantitative commitments. The first is the *renewable share*: across the duration of the continual learning sweep, the share of energy delivered to the AI accelerators that originates from renewable-native sources (locally generated photovoltaic plus battery storage discharge attributable to locally generated photovoltaic, as opposed to grid imports) is required to exceed 80 percent, as measured by the unified measurement plane and as attested by the green-compute certificate. The second commitment is the *JCT inflation cap*: the total job completion time of the sweep is required to be within 130 percent of the JCT that the same sweep would have on an equivalent baseline running on commercial cloud GPUs without renewable constraints; the cap captures the realistic cost of operating under intermittent renewable energy and is calibrated to be tight enough to be operationally meaningful and loose enough to be physically achievable. The third commitment is the *in vivo demonstration of the equivalence principle*: across the duration of the sweep,

the orchestrator must dispatch energy through both sub-modes of the time lever (battery storage and compute acceleration on deferrable backlog) and must publish, in the certificate, the realised partition between the two sub-modes; the partition is one of the central scientific outputs of the testbed, because it is the first measured datapoint on the silicon–lithium isopreference frontier introduced in Section IV-E.

D. AI-RAN as the second application load

The continual learning workload is the primary load of the testbed but is not the only one. A second application load is provided by an AI-RAN node, deployed at one of the consortium sites and integrated into the federation as an additional class of compute endpoint with its own QoS contract and its own envelope shape. The AI-RAN node serves a real or emulated radio service whose latency budget is at the millisecond scale and which, as discussed in Section III-E, is the tightest stress test of the envelope analysis. On the same hardware, in slots where the radio scheduler releases the accelerator, the AI-RAN node accepts general-purpose AI workloads coming from the federation through the standard nomadic admission path; these workloads see a shrunk envelope because of the coexistence with the radio function,

and the testbed quantifies the shrinkage empirically against the predictions of the framework.

The role of the AI-RAN node in the testbed is therefore twofold. On the one hand, it is a deployment site at the very edge of the eligibility envelope, and its results validate (or falsify) the prediction that the framework remains operational at this edge. On the other hand, it is a representative of a workload class (latency-bounded radio-adjacent AI inference) that is qualitatively different from the continual learning workload, and its inclusion in the testbed broadens the empirical base of the validation beyond a single workload type. The combination of the two loads, served by the same orchestrator and certified by the same green-compute mechanism, is what makes the testbed a genuine end-to-end demonstration rather than a single-workload pilot.

E. What the testbed empirically settles

The federated testbed is sized to settle, with measurable evidence, four claims that the framework makes and that today have no empirical anchor. First, that a workload of contemporary operational scale (Llama-class continual learning) can in fact be served by an inter-site federation operating under renewable constraints, at a cost in JCT that remains within an order of magnitude of the cloud baseline. Second, that the storage–compute equivalence principle is not only a theoretical observation but a deployable orchestration mechanism: the same surplus can be absorbed through either sub-mode of the time lever, and the partition between the two sub-modes is a controllable degree of freedom rather than a fixed property of the infrastructure. Third, that the verifiable green-compute certificate is producible end to end and is auditable by a third party: a regulator, an auditor, or the workload owner can verify the renewable share and the JCT trajectory from the certificate alone, without privileged access to the operator’s internal telemetry. Fourth, that the framework extends, without structural modification, to a qualitatively different workload class (AI-RAN) deployed at the tightest edge of the eligibility envelope.

These four claims, settled empirically, are the substance of the GridAI commitment to the funding agency, to the consortium’s industrial stakeholders, and to the wider European policy and research community now engaged in the build-out of the renewable-native energy and digital infrastructure of the next decade. The testbed is not a demonstration that the nomadic paradigm replaces the HVDC paradigm; the eligibility test makes the two paradigms complementary by construction, and the share of European curtailment that the nomadic envelope can credibly address is a fraction of the total HVDC-targeted volume which, while small in relative terms, is large in absolute terms and large enough to justify a dedicated infrastructure. The testbed is, instead, a demonstration that the nomadic paradigm is operationally credible at the scale at which it would need to be deployed, with the verification machinery that would be required for it to be procured by public and private actors. This is, in our view, the minimum empirical commitment that justifies treating the nomadic AI

compute paradigm as a policy option rather than as a research speculation.

VII. OPEN CHALLENGES

A. What the framework does not yet settle

The framework articulated in the preceding sections is, in our view, sufficient to elevate the nomadic AI compute paradigm from a slogan to an operational design discipline. It is not, however, sufficient to answer every question that the paradigm raises, and we close this paper by stating openly the questions that remain open and on which we invite the broader community to contribute.

The first open challenge is the empirical calibration of the deferrability axis of the envelope at population scale. Today we have a handful of well-characterised workload classes (live conversational inference, interactive fine-tuning, overnight continual learning, periodic batch sweeps) whose deferrability profiles are reasonably well understood; we do not yet have a representative cross-sectional measurement of the full distribution of deferrability across the AI workloads that traverse a commercial cloud or a research infrastructure. The cost functional and the storage–compute equivalence principle are operationally meaningful only to the extent that this distribution is known, because the partition between battery storage and compute over-provisioning at a site depends on the volume of deferrable workloads that the site can expect to see over a year. Producing a credible measurement of this distribution, on either a research or a commercial dataset, is a contribution that we cannot make alone.

The second open challenge is the formalisation of the eligibility test as a continuous metric rather than a binary admission criterion. The framework currently treats eligibility as a sharp boundary between the workloads to which the nomadic envelope applies and the workloads (FullSync data-parallel training of frontier models) that must be served by co-located large facilities. In reality, the boundary is gradual: a workload that requires moderate inter-site bandwidth on a tightly coupled fabric can be served at a higher latency and at a worse JCT on a looser fabric, and the cost functional should in principle quantify this degradation rather than rejecting the workload outright. Producing a continuous eligibility metric, calibrated against the algorithmic regime variable ρ , is a natural extension of the framework that we anticipate working on in the next phase.

The third open challenge is the integration of the GridAI orchestrator with the regulatory framework of compute-as-flexibility under development at the European and national level. The Clean Energy Package, the ARERA framework on TSO/DSO coordination, the prospective evolution of the ancillary-services market, and the on-going regulatory work on DER aggregation all interact with the nomadic compute paradigm in ways that the framework has only partially addressed. A credible integration requires participation by the relevant regulator, by the system operator, and by the cloud and compute providers, on a timescale that is longer than a single research project; the GridAI project commits to producing

the technical and methodological inputs to this regulatory conversation, but the conversation itself is necessarily a wider one.

The fourth open challenge is the verifiability of the green-compute certificate under adversarial assumptions. The current candidate design rests on a site-level cryptographic root-of-trust combined with a tamper-evident verifiable log; this is a defensible starting point but is not the end of the story, because a determined adversary can in principle compromise either component, and the verification properties of the certificate degrade gracefully in some failure modes and ungracefully in others. A systematic security analysis of the certificate format, ideally produced in conjunction with the standards body that will host the format, is a deliverable that we have not committed to inside the project scope but that we identify as a necessary follow-up.

B. The position of GridAI in the European landscape

The GridAI project is being built in a European context in which several large initiatives, with substantial public and private funding, are addressing closely related problems from different angles. The transcontinental HVDC build-out (SuedLink in Germany, the Tyrrhenian Link and ELMED in Italy and the central Mediterranean, the North Sea bootstraps in the United Kingdom) is the most visible expression of the electron-transport paradigm and is, as discussed, complementary rather than competitive with the nomadic compute paradigm; the silicon–lithium isopreference frontier of Section IV-E is, among other things, a quantitative interface between the two paradigms, and we expect it to be the natural axis along which their coexistence is negotiated.

On the compute side, the European Union has been deploying, through the EuroHPC programme, a fleet of national high-performance computing sites that are increasingly being used for AI workloads; these sites are large and co-located with substantial energy infrastructure, and they sit at the high-end boundary of the four-scale taxonomy we have proposed. The integration of the nomadic compute paradigm with the EuroHPC fleet is, in our view, one of the most consequential cross-cutting conversations that the European research community should be having; it is the conversation that determines whether the next generation of European compute infrastructure is built as a small number of large co-located facilities, as a federated nomadic infrastructure across many smaller energy-proximate sites, or (most likely) as a combination of the two with the partition explicitly designed rather than left to emerge from independent procurement decisions.

On the wireless side, the AI-RAN Alliance and the broader 6G research programme have produced a substantial body of work on the colocation of AI compute and radio function at the network edge, much of which is directly relevant to the boundary-case analysis of Section III-E. The cross-fertilisation between the AI-RAN community and the renewable-native compute community has been limited so far, and we believe the boundary case framing of GridAI is one of the natural

vehicles through which the two communities can be brought into the same conversation.

C. A note on European collaboration

The empirical validation that GridAI is committing to deliver over its three-year horizon will be, by construction, anchored on Italian sites and on Italian grid and workload data; this is the right scope for a single project of the size proposed, but it is insufficient for the European-scale conclusions that the framework is designed to inform. A separate companion document *GridAI – Call for European Collaboration* accompanies this paper and articulates three concrete tracks (cross-validation of the simulator on non-Italian grid data, cross-deployment of the federated testbed across European sites via GARR/GÉANT, and IETF standardisation around green-computing monitoring and certification) along which the GridAI consortium invites engagement.

VIII. CONCLUSION

AI orchestration today operates under three implicit assumptions whose joint effect is to make the renewable-native compute paradigm invisible. The first assumption is that every AI workload is hard-real-time; the second is that workloads are nailed to the site at which they were admitted; the third is that battery storage is the only form of energy flexibility available to a compute infrastructure. All three assumptions are false under verification, and the cumulative cost of acting as though they were true is the structural mismatch between the geography of renewable generation and the geography of compute consumption that the European HVDC build-out is now being asked to absorb.

The framework developed in this paper replaces the three assumptions with physical and measurable properties of the nomadic envelope. Workloads have explicit deferrability profiles, on which the orchestrator acts as a first-class scheduling input. Workloads have explicit spatial-mobility properties, expressed as the joule and time cost of relocating their state across the inter-site fabric, and a substantial fraction of contemporary AI workloads sit comfortably inside the feasibility region of this mobility under realistic renewable windows. Flexibility has two interchangeable physical embodiments, lithium and silicon, related by an equivalence principle that makes the partition between the two a controllable design variable and the central object of the silicon–lithium isopreference frontier.

The position taken by the GridAI consortium is not that the nomadic compute paradigm replaces the electron-transport paradigm. The eligibility test of the framework is explicit about the workloads that fall outside its scope, and these workloads will continue to drive the HVDC build-out at continental scale. The position is, rather, that the partition between the two paradigms is today decided by default rather than by design, and that the methodological tools needed to decide it by design (the cost functional, the simulator, the green-compute certificate, the multi-scale infrastructure taxonomy) are within the reach of a focused three-year research programme calibrated on the Italian electrical and computational landscape.

The empirical commitment that the consortium attaches to this position is the federated end-to-end demonstration of Section VI: a Llama-class continual learning workload, federated across at least three energy-proximate micro-clusters, with a renewable share above 80 percent, a JCT within 130 percent of a cloud baseline, an explicit in-vivo demonstration of the storage–compute equivalence principle, and a third-party-auditable green-compute certificate covering the entire trajectory. The commitment is qualified rather than absolute, because the framework is honest about the algorithmic regimes under which it operates; this qualification is the precondition for the commitment to be technically defensible and for the larger conversation that the paper is intended to open to be a productive one.

Following the sun and the wind, the way the GridAI framework articulates it, is not a slogan. It is a multi-scale infrastructure design, a multi-lever cost functional, an equivalence principle, an eligibility test, a verifiable certification format, and an empirical commitment at a defined horizon, with each component designed to be replaceable, extensible, and standardisable as the European and global conversation on the renewable-native digital infrastructure matures over the coming decade. The invitation to that conversation is the substance of this paper, and the GridAI consortium signs it together with the industrial partners who co-author it: a scientific bet, formulated as carefully as we know how to formulate it, on the proposition that the right answer to the collision between AI and the energy transition is to move bits, not electrons, every time the workload’s nomadic envelope allows it.

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